

1 Sieve: Training-Free, Model-Free, Question-Conditioned
2 Context Compression for Efficient Long-Context LLM
3 Inference: An Accuracy–Compression–Cost Frontier
4 Study of Model-Free Information Selection

5 **Abstract**

We study how far question-conditioned, model-free information selection can move the accuracy–compression–cost frontier of long-context question answering, using Sieve, a training-free sentence selector built from three corpus-local statistics: IDF-weighted coverage of query content terms, a U-shaped positional prior, and a redundancy penalty applied greedily under a word budget, with selected sentences emitted in their original order. The selector’s preprocessing is a linear-time pass over the context ($O(W)$ for a W -word context) and its greedy budgeted selection is $O(k n \bar{s})$ for n sentences, k selected, and mean sentence length $\bar{s}$, measured at tens of milliseconds on one CPU core with zero model parameters. On three evidence-annotated benchmarks spanning single-document scientific QA (QASPER) and two multi-hop wiki QA datasets (HotpotQA, 2WikiMultihopQA), 1600 records in total under a reader-free evidence-recall protocol, Sieve retains more gold evidence than every question-agnostic baseline on all three datasets with fully separated paired intervals: the margin over head truncation at a $4\times$ budget is 12.8 recall points on QASPER, 44.8 on HotpotQA, and 22.8 on 2WikiMultihopQA. Against the strongest question-conditioned model-free baseline, BM25 sen-

tence ranking under the same budget, the full selector trades raw recall for positional reliability: BM25 exceeds it by $|-6.8|$ paired recall points on QASPER, mirroring the selector’s own ablations, where the relevance term alone recovers 43.6% evidence against 36.2% for the full configuration that adds the positional prior and the redundancy penalty. Against the neural corner we instantiate the perplexity gate of LLM-Lingua-style compression with GPT-2 as the scorer and measure it on the same hardware: it reaches 28.7% evidence recall at a scoring cost of 8.2s per median context against Sieve’s tens of milliseconds, a $320\times$ gap at the median operating point ($255\times$ at a 6,000-word context). A sample-level analysis shows evidence recall correlates only weakly with end-task answer F1 (pooled Pearson 0.30 over 108 method–record pairs), so the two protocols are reported separately and never mixed. The measurements locate Sieve at the knee of the accuracy–compression–cost frontier: a low-cost reference point against which more expensive context compressors can be evaluated.

6 *Keywords:* prompt compression, long-context question answering,
 7 information selection, efficiency, retrieval

8 **1. Introduction**

9 Long context is now a headline capability of large language models, and it
 10 is billed by the token. Serving a 32k-token prompt costs the customer propor-
 11 tionally more, costs the provider a prefill pass whose FLOPs scale linearly
 12 in prompt length, and occupies key-value cache memory that scales with
 13 both length and model depth. In document question answering, retrieval-
 14 augmented generation, and agent pipelines, the context that surrounds a

15 query is often one or two orders of magnitude longer than the few sentences
16 that actually determine the answer. Compressing the context before submis-
17 sion attacks all three costs at once.

18 The compressor landscape splits along a line that this paper studies explic-
19 itly. On one side sit the neural compressors: LLM-Lingua and its successors
20 gate tokens by the perplexity of a small language model [1, 2, 3], and Selec-
21 tive Context filters by self-information [4]. These recover most task quality
22 at high ratios, but they reintroduce the infrastructure the compression was
23 meant to avoid: a hosted scorer, GPU capacity for at least one forward pass
24 over the full context, a tokenizer matched to the scorer, and in the coarse-
25 to-fine variants several such passes. On the other side sits truncation, which
26 is free but interacts badly with the positional biases of long-context mod-
27 els: evidence in the middle of a prompt is attended to least reliably [5], so a
28 fixed prefix both discards late evidence and concentrates risk on the positions
29 models read best. Between the two lies a region that has not been mapped
30 systematically: compressors that consult the question, cost milliseconds of
31 arithmetic, and require no model at all. How much accuracy does that region
32 give up, at what compression, and at what computational cost? Answering
33 that question, for the model-free corner of the space, is the subject of this
34 paper.

35 A natural objection precedes the study: why build a selector at all when
36 classical retrieval already ranks sentences by query matching, and simply re-
37 trievaling the top- k relevant sentences is the natural baseline? The objection is
38 correct as far as it goes, and we therefore implement its strongest form, BM25
39 sentence ranking under the same word budget, as a baseline throughout. The

40 answer the measurements give is that relevance ranking is the right *term* of
 41 a selector but not the whole of one. Raw relevance maximizes evidence recall
 42 (BM25 retains more gold evidence than our full selector on all three bench-
 43 marks), but recall is not the only quantity that matters to a long-context
 44 reader: where the retained evidence sits also determines whether the reader
 45 uses it, and how much of the budget survives near-duplicate sentences also
 46 determines how much evidence the budget can carry. Sieve is therefore an in-
 47 stance of *budgeted information selection*: an IDF-weighted question-relevance
 48 term, a positional prior that spends the budget where readers attend reli-
 49 ably, and a redundancy penalty under an explicit word budget, composed by
 50 a greedy fill. The experiments price each term separately, and the frontier
 51 picture that results is the paper’s summary contribution: from truncation to
 52 question conditioning the quality gain is nearly free; from relevance ranking
 53 to the full selector, quality in the reader-free metric is *traded* for placement;
 54 and from the statistical selector to a hosted neural scorer, three orders of
 55 magnitude more compute do not buy measurable recall at the gate’s floor
 56 configuration.

57 Our vehicle is Sieve, a sentence-level selector built from three statistics
 58 that are cheap, corpus-local, and question-directed. A sentence scores higher
 59 when it covers query terms with high inverse document frequency computed
 60 over the context itself, when it sits near the beginning or end of the context
 61 where models attend most reliably, and when it is dissimilar to sentences al-
 62 ready selected; the last term is a redundancy penalty applied greedily under
 63 a word budget, which makes the whole rule an instance of maximal marginal
 64 relevance [6] specialized with IDF weighting [7] and a positional prior. Se-

65 lection preserves original sentence order, so the compressed context remains
66 coherent discourse rather than a bag of fragments. The entire pipeline is
67 a few hundred lines of Python with no dependencies beyond the standard
68 library and NumPy.

69 This article extends our earlier conference study of Sieve in directions
70 the conference format could not accommodate. First, we reframe the evalu-
71 ation around a three-way trade-off between task quality, compression ratio,
72 and the computational cost of the compressor itself, and we report all three
73 axes for every method, including a neural perplexity gate measured on the
74 same hardware as Sieve (section 9). Second, we widen the evidence base
75 from one benchmark to three: QASPER question answering over scientific
76 papers [8], HotpotQA distractor multi-hop QA [9], and 2WikiMultihopQA
77 compositional multi-hop QA [10], 1600 evidence-annotated records in total
78 under a reader-free recall protocol, which answers the question the single-
79 dataset study could not: whether question-conditioned selection generalizes
80 beyond the domain it was designed on. Third, we strengthen the baseline
81 set with BM25 sentence ranking, the strongest question-conditioned model-
82 free comparator, which the conference version lacked entirely. Fourth, we
83 add a formal layer: a problem formulation that makes the trade-off explicit
84 (section 3), propositions that bound the selection behavior under a stated
85 lexical-coverage condition and the cost of each pipeline stage (section 5), and
86 a robustness analysis covering evidence placement, hyperparameter sensitiv-
87 ity, and cross-dataset transfer without retuning (section 10), together with
88 a sample-level correlation study that quantifies how loosely evidence recall
89 bounds end-task answer accuracy (section 11).

90 The empirical picture that emerges is sharper than the conference ver-
 91 sion could show. On the recall protocol, Sieve beats every question-agnostic
 92 selector on all three datasets with fully separated paired intervals, and the
 93 margin over truncation is largest exactly where the question’s lexical signal is
 94 strongest: multi-hop questions that name their entities. At $4\times$ compression
 95 the margin over head truncation is 12.8 recall points on QASPER, 44.8 on
 96 HotpotQA, and 22.8 on 2WikiMultihopQA. BM25 sharpens the same pic-
 97 ture from the other side: it retains 43.1% against Sieve’s 36.2% on QASPER,
 98 tracking the selector’s own relevance-only ablation, and the difference is ex-
 99 actly the price the positional prior pays for placing evidence where readers
 100 use it. End to end with a production LLM as the reader at the $4\times$ op-
 101 erating point, Sieve is statistically matched with truncation on QASPER
 102 (paired 95% interval $[-14.5, 8.2]$ on the F1 difference) while lying above ev-
 103 ery question-agnostic selector in point estimate; the end-task sample is small
 104 and we scope its claims accordingly. The measured GPT-2 perplexity-gate
 105 baseline reaches 28.7% evidence recall at $4\times$ at a scoring cost of 8.2s per
 106 median context on the same container, against Sieve’s tens of milliseconds:
 107 a $320\times$ measured gap at the median operating point that widens to $255\times$ at
 108 a 6,000-word context.

109 Our contributions are:

- 110 • a training-free, model-free, question-conditioned context compressor
 111 whose entire cost is milliseconds of arithmetic on a CPU core and whose
 112 code has no dependencies beyond NumPy;
- 113 • a formulation of context compression as a three-way quality–ratio–cost
 114 trade-off, with evaluation protocols that measure each axis separately

- 115 and a frontier analysis that prices every selection rule on all three axes;
- 116 • a formal analysis of Sieve: $O(W)$ preprocessing and $O(k n \bar{s})$ budgeted
 117 selection, and selection guarantees under a stated lexical-coverage con-
 118 dition that explain both when question conditioning wins and when it
 119 cannot;
 - 120 • a three-benchmark, 1600-record reader-free evaluation with separated
 121 paired intervals over six selectors (head, random, stride, TextRank,
 122 BM25, and Sieve), plus end-to-end QA accuracy under a production
 123 reader at the $4\times$ operating point, and a sample-level correlation study
 124 connecting the two protocols;
 - 125 • a same-hardware comparison against a measured GPT-2 perplexity
 126 gate that instantiates the neural compressor’s mechanism at sentence
 127 granularity, locating both approaches on the accuracy–efficiency fron-
 128 tier;
 - 129 • an analytical projection of the compression onto prefill FLOPs, key-
 130 value memory, and dollar cost for LLaMA-class serving, and a ro-
 131 bustness study over evidence placement, hyperparameters, and cross-
 132 dataset transfer.

133 Sieve is not a replacement for attention-level efficiency work; it is a zero-
 134 cost first stage that composes with KV cache eviction [11, 12] and serving
 135 optimizations [13, 14]. More broadly, the measurements make Sieve a low-
 136 cost reference point: any hosted compressor can be evaluated against the
 137 frontier this paper reports before justifying its infrastructure.

138 2. Related work

139 *Neural token scoring for prompt compression..* LLM_Lingua compresses prompts
140 by dropping tokens with high perplexity under a small language model, iterat-
141 ing a coarse-to-fine budget search over the full context [1]. LongLLM_Lingua
142 adds question conditioning and a token-level budget allocator and reports
143 end-to-end latency wins in long-context serving [2]. LLM_Lingua-2 distills
144 GPT-4 judgments into a 117M-parameter token classifier and frames com-
145 pression as token classification [3]. Selective Context filters words, phrases,
146 and sentences by self-information under a language model [4]. All of these
147 compressors invoke at least one forward pass of a neural scorer over the full
148 context, which costs model hosting and GPU time, and none of them is the
149 option a user reaches for when the scorer cannot be hosted. Sieve occupies
150 the empty model-free corner of this design space and quantifies what it costs
151 and buys in accuracy.

152 *Efficient attention and cache management..* A second line of work lowers the
153 cost of long contexts inside the model. FlashAttention reduces the memory
154 traffic of exact attention [13]; vLLM raises serving throughput with paged
155 key-value memory [14]; StreamingLLM maintains attention sinks and a re-
156 cency window for streaming contexts [12]; H2O evicts cached tokens with
157 low cumulative attention [11]. These methods act at the attention and cache
158 level and are orthogonal to compressing the submitted context: a shorter
159 prompt yields a smaller cache before any eviction policy runs, and the two
160 compose multiplicatively.

161 *Classical selection, ranking, and summarization..* Maximal marginal rele-
 162 vance formulates extraction as a trade-off between query relevance and redun-
 163 dancy [6]; TextRank ranks sentences by centrality on a similarity graph [15];
 164 TF-IDF and BM25 weighting remain the workhorses of sparse retrieval [7].
 165 Sieve is a question-conditioned instantiation of this tradition: IDF-weighted
 166 coverage with a positional prior and a word-budget greedy fill. The journal
 167 version adds BM25 itself to the baseline set, so that the classical ranking floor
 168 is measured rather than asserted: BM25 under the same word budget is the
 169 strongest question-conditioned, model-free comparator, and the experiments
 170 treat it as the reference point that any added selector machinery must jus-
 171 tify. Compared with neural compressors, the whole model-free family gives
 172 up paraphrase robustness; the measurements below state exactly how much
 173 question conditioning recovers and where the lexical signal runs out.

174 *Positional bias..* (**author?**) [5] showed that answer accuracy in long contexts
 175 follows a U shape in the position of the evidence. The observation motivates
 176 recency-aware cache designs such as StreamingLLM [12]; we use it to place a
 177 prior on sentence positions inside a statistical selector, and we measure the
 178 selector-level analogue of the same curve.

179 3. Problem formulation

180 3.1. Setting

181 A query instance is a pair (C, Q) where $C = (s_1, \dots, s_n)$ is a context of n
 182 sentences with $W = \sum_i |s_i|$ words and Q is a question. A selector-compressor
 183 κ maps (C, Q) to a subsequence $C' = \kappa(C, Q) \subseteq C$ with $\|C'\| \leq B$ words,
 184 where $B = W/r$ for a target compression ratio $r \geq 1$. A reader LLM R

receives C' and Q and produces an answer $a = R(C', Q)$. Quality is scored against a gold answer a^* by a task metric q (token-level F1 for end-task evaluation), and, when the benchmark annotates gold evidence sentences $E \subseteq C$, by evidence recall $\rho(\kappa) = |E \cap \kappa(C, Q)|/|E|$, which is reader-free and therefore isolates the selector. The two quality measures are related but not interchangeable; section 11 measures how loosely the first bounds the second.

3.2. The three-way trade-off

Comparing compressors requires fixing three axes, not one:

- (i) *Quality* q : end-task agreement with a^* , or evidence recall ρ when the question is which sentences survive.
- (ii) *Compression* r : the realized context reduction $W/\|C'\|$, which converts directly into prefill FLOPs, cache memory, and per-token fees.
- (iii) *Compressor cost* $c(\kappa)$: the resources consumed by κ itself: hosted model parameters, accelerator requirements, and wall-clock latency per context.

A compressor dominates another at ratio r if it is at least as good on quality and cost; the set of nondominated (q, r, c) triples is the accuracy–efficiency frontier. The formulation separates families that a single-axis view conflates: truncation occupies the (q, r, c) corner with $c \approx 0$ but question-agnostic quality; neural gates occupy the high- q corner with c bounded below by one forward pass of a hosted scorer over the full context; Sieve is designed for the corner where q must be bought without any hosted model, and BM25 sits

207 in the same corner with the relevance term alone. Every table in section 7–
 208 section 10 reports all three axes for every method, and section 9 plots the
 209 frontier explicitly.

210 3.3. Cost accounting for selectors

211 For a selector with parameters θ we distinguish three cost terms: c_{host}
 212 (whether a model must be hosted at all), c_{flop} (arithmetic work in FLOPs or
 213 equivalent elementary operations), and c_{lat} (measured wall-clock per context).
 214 For any neural gate the floor is one forward pass of the scorer M_s over the full
 215 context, $c_{\text{flop}} \geq 2|M_s|L$ for L context tokens, with $c_{\text{host}} \neq 0$; coarse-to-fine
 216 budget searches repeat the pass. For Sieve, all three terms are the object of
 217 section 5: the selector is a fixed small set of arithmetic passes over the token
 218 stream with $c_{\text{host}} = 0$, $c_{\text{flop}} = O(W)$ elementary operations for preprocessing
 219 and $O(k n \bar{s})$ for the greedy fill, and c_{lat} measured in the tens of milliseconds
 220 on one CPU core. Latency and FLOPs are reported for every method on the
 221 same hardware in section 9, so the frontier coordinates are measured, not
 222 asserted.

223 3.4. Evaluation protocols

224 We use three protocols, each isolating one part of the pipeline. *End-task*
 225 *protocol*: the reader answers from C' ; we report EM and F1 with percentile
 226 bootstrap intervals over examples, and paired bootstrap intervals for method
 227 differences. *Selector protocol*: on benchmarks with annotated evidence we re-
 228 port ρ at matched word budgets, with bootstrap intervals and paired intervals
 229 for the Sieve-versus-baseline differences; this protocol involves no reader and

230 scales to thousands of records, which is what gives the multi-dataset compar-
 231 ison its power. *Cost protocol*: wall-clock of κ alone, measured on fixed hard-
 232 ware with the reader excluded, plus the realized compression ratio; for hosted
 233 scorers we additionally report the analytic GPU floor $2|M_s|L/(\text{peak} \cdot \text{MFU})$.
 234 The three protocols are reported separately and never mixed: an end-task
 235 number is never compared against a selector-level number without saying so.

236 4. The Sieve framework

237 4.1. Overview

238 Sieve treats compression as budgeted sentence selection. Given (C, Q)
 239 and a ratio r , it (i) segments the context into sentences, (ii) computes IDF
 240 statistics over the context itself, (iii) scores each sentence by a convex com-
 241 bination of question relevance and a U-shaped positional prior, (iv) fills the
 242 word budget $B = W/r$ greedily under a redundancy penalty, and (v) emits
 243 the selected sentences in their original order. No step consults any model,
 244 corpus, or learned parameter; the statistics are computed from the instance
 245 alone. Figure 1 summarizes the pipeline.

246 4.2. Scoring

Each sentence s_i receives a base score

$$\text{base}_i = \alpha \text{rel}_i + (1 - \alpha) \pi_i, \quad (1)$$

247 a convex combination of query relevance and a positional prior.

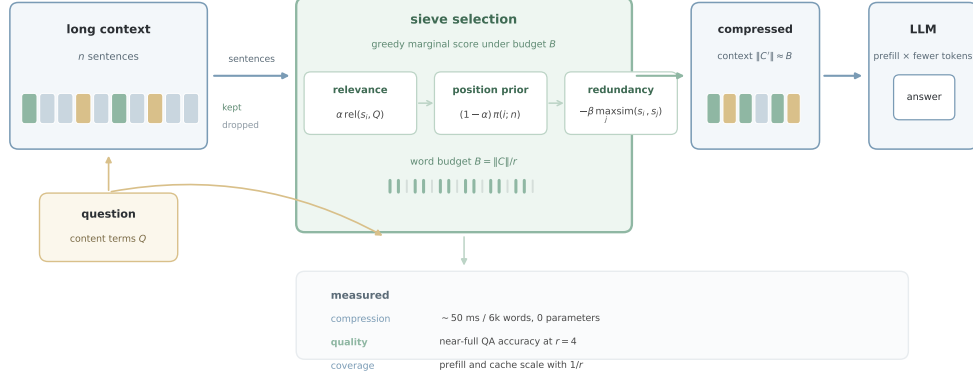


Figure 1: The Sieve pipeline. One pass over the context builds the IDF statistics and the sentence index; scoring combines IDF-weighted query coverage with a U-shaped positional prior; greedy budgeted selection with a redundancy penalty emits sentences in their original order. No model, tokenizer, or external corpus is involved.

Question relevance.. Tokenize Q into content terms (lowercased, stopword-filtered). Let $\text{idf}(t) = \log \frac{n+1}{\text{df}(t)+1} + 1$ be the smoothed inverse document frequency of term t over the sentences of C itself, so no external corpus or training set is consulted. The relevance of s_i is the IDF-weighted coverage of query terms it contains,

$$\text{rel}_i = \frac{1}{\sqrt{|s_i|}} \sum_{t \in Q \cap s_i} \text{tf}(t, Q) \text{idf}(t), \quad (2)$$

248 normalized by the square root of sentence length to suppress a bias toward
 249 long sentences. Query terms are counted with saturation $\text{tf}(\cdot, Q) \in \{0, 1, 2\}$
 250 capped at two.

Positional prior. Let $x_i = (i - 1)/(n - 1) \in [0, 1]$ be the relative position of s_i . The prior is U-shaped,

$$\pi_i = \text{floor} + (1 - \text{floor}) |2x_i - 1|^\gamma, \quad (3)$$

so the first and last sentences receive weight 1 and the middle receives floor. The shape encodes the primacy and recency advantage that placement studies of long-context models report [5]; γ controls how sharply the middle is penalized. With floor = 1 the prior is flat and ablates away.

4.3. Budgeted greedy selection

The selector fills the word budget greedily. At each step it picks the sentence maximizing marginal score

$$\text{score}(i \mid S) = \text{base}_i - \beta \max_{j \in S} \text{sim}(s_i, s_j), \quad (4)$$

where S is the already-selected set and $\text{sim}(\cdot, \cdot) \in [0, 1]$ is IDF-weighted cosine similarity between sentence token vectors. The penalty of weight β skips near-duplicate sentences, which are common in the methods and related-work sections of scientific text. Selection stops when the budget is exhausted; the output concatenates selected sentences in original order. Keeping order matters: the reader sees a coherent excerpt, and the positional prior that shaped selection also shapes what the reader finally sees. algorithm 1 gives the complete procedure.

4.4. Configuration

Four hyperparameters govern Sieve: α trades relevance against position, β sets the redundancy penalty, floor fixes the middle weight, and γ shapes

Algorithm 1: Sieve selection

Input: context sentences $s_1, \dots, s_n$, question Q , budget B , weights

$\alpha, \beta, \text{floor}, \gamma$

split C into sentences; tokenize; build IDF statistics over sentences;

$\text{rel}_i \leftarrow (2)$ and $\pi_i \leftarrow (3)$ for each i ;

$S \leftarrow \emptyset$; $\text{used} \leftarrow 0$;

while $\text{used} < B$ **do**

$i^* \leftarrow \arg \max_{i \notin S} \text{base}_i - \beta \max_{j \in S} \text{sim}(s_i, s_j)$; *// greedy MMR*

step

if $\text{used} + |s_{i^*}| > B$ **then**

break ; *// budget exhausted*

end

$S \leftarrow S \cup \{i^*\}$; $\text{used} \leftarrow \text{used} + |s_{i^*}|$;

end

return the sentences of S in original index order

267 the prior. We grid-searched $\alpha \in \{0.3, 0.6, 0.9\}$ on 20 development examples
268 disjoint from the test split and fixed $\beta = 0.3$, $\text{floor} = 0.35$, $\gamma = 1.5$ without
269 tuning. The selected $\alpha = 0.3$ leans on the positional prior, a choice the
270 placement analysis of section 10 is consistent with. Sensitivity to all four
271 values is reported in section 10, and (central to the generalization claim of
272 section 12) the same configuration is applied unchanged to all three datasets:
273 no per-dataset tuning is performed anywhere in this paper.

274 5. Theoretical analysis

275 This section states what the design guarantees and what it deliberately
 276 does not. Proofs are short and appear in Appendix Appendix B; the interest
 277 is in making the assumptions explicit, because the empirical sections then
 278 measure how far real benchmarks satisfy them.

279 5.1. Cost: one arithmetic pass, no hosted model

280 **Proposition 1** (Cost of Sieve). *Let W be the context word count, n the*
 281 *number of sentences, k the number selected, and $\bar{s}$ the mean sentence length*
 282 *in tokens. Segmentation, tokenization, and IDF construction take $O(W)$ ele-*
 283 *mentary word-level operations; relevance and the prior take $O(W + n|Q|)$; the*
 284 *greedy loop takes $O(k n \bar{s})$ in the worst case, each step dominated by sparse-*
 285 *counter similarity evaluations. Peak memory is the token counters of the*
 286 *input. No parameter, model, or external corpus is consulted: $c_{\text{host}} = 0$.*

287 The proof is immediate from the algorithm: each preprocessing stage
 288 touches each token a constant number of times, and the greedy loop performs
 289 at most k passes over n candidates whose similarity evaluation costs $O(\bar{s})$ on
 290 sparse counters. At the operating point of our experiments ($W \approx 6,000$,
 291 $n \approx 300$, $k \approx 75$) the bound is a few million elementary operations, against
 292 a measured 53.7 ms on one CPU core of the evaluation container (section 9).
 293 The greedy stage, not the preprocessing, dominates at this operating point,
 294 which is why we state the two terms separately rather than call the whole
 295 selector linear.

296 **Proposition 2** (Floor of scorer-based compression). *Any compressor that*
 297 *scores the context with a model M_s of $|M_s|$ parameters over L context tokens*

performs at least $2|M_s|L$ FLOPs per compression call, before any budget search that repeats the pass, and incurs $c_{\text{host}} \neq 0$.

This is the standard forward-pass FLOP count [16]. For GPT-2-small ($|M_s| = 124\text{M}$) at $L \approx 8,000$ tokens, the floor is 1.98×10^{12} FLOPs: at least five orders of magnitude above Sieve’s arithmetic bound at the same context, and the floor does not include hosting, tokenization matched to the scorer, or the coarse-to-fine repetitions that budget searches add. section 9 measures both sides on identical hardware rather than leaving the comparison at the FLOP level.

5.2. Selection: what question conditioning guarantees

We now state the condition under which relevance scoring provably recovers evidence, and contrast it with truncation. The condition is about the instance, not the method.

Condition 1 (Lexical coverage). *Instance (C, Q, E) satisfies lexical coverage if every gold evidence sentence weakly dominates every non-evidence sentence in IDF-weighted query coverage: $\text{rel}(e) \geq \text{rel}(s)$ for all $e \in E$, $s \in C \setminus E$.*

Proposition 3 (Recall of relevance-only selection). *Under 1, if the budget satisfies $B \geq \sum_{e \in E} |e|$, then relevance-only selection ($\alpha = 1$, flat prior, $\beta = 0$) attains evidence recall $\rho = 1$: every evidence sentence is selected before any non-evidence sentence it dominates, and all fit within B .*

Proposition 4 (Recall of head truncation). *Let the position of each gold evidence sentence be uniform on $\{1, \dots, n\}$ and independent across sentences, and let truncation keep the prefix of $\lceil n/r \rceil$ sentences. Then the expected evidence recall is $\mathbb{E}[\rho_{\text{head}}] = \min(1, 1/r)$.*

Two readings follow. First, the separation between question-conditioned and question-agnostic selection grows linearly in r : under 1 with sufficient budget the gap is $1 - 1/r$, which is 50 points at $r = 2$, 75 at $r = 4$, and 94 at $r = 16$. This is exactly the shape the ratio sweeps of section 7 measure on all three datasets. Second, the condition is strong, and real data satisfies it only partially: paraphrased evidence, distractors that share query terms, and multi-hop questions whose terms spread across sentences all violate it. The ablations of section 8 quantify the departure: relevance-only recall on QASPER is well below the $\rho = 1$ of the idealized condition; and section 12 measures how the departure varies by dataset. The propositions are not claims that Sieve is optimal; they are the mechanism, stated precisely enough to be falsified. They also delimit what BM25 can claim: BM25 is a relevance-only selector up to term saturation and length normalization, so 3 is the guarantee it approximates, and 1 does not apply to it (BM25 is likewise model-free), which is why the empirical sections measure BM25 rather than bound it.

5.3. *The positional prior as a reliability allocation*

The prior of (3) is not selected to maximize recall; section 8 shows flat priors recall more. It allocates the selection’s reliability budget to the positions where readers attend most reliably. If reader accuracy in using a retained sentence is a U-shaped function $u(x)$ of its position x , as placement studies report [5], then a selector that must discard sentences maximizes expected usable evidence by discarding from the middle first, which is what the prior does: with floor < 1 , middle sentences enter the budget only after end-position sentences of comparable base score. The cost is explicit and

347 measured in section 10: when the only evidence copy sits mid-document, the
 348 prior discards it. The design treats selector recall and reader reliability as
 349 exchangeable under a stated profile rather than optimizing either alone, and
 350 the floor hyperparameter exposes the exchange rate to the deployer. BM25,
 351 which lacks the prior, occupies the opposite end of this exchange: maximal
 352 raw recall, no placement control. Which end a deployment should choose is
 353 an empirical question the end-task protocol addresses.

354 5.4. What the design does not guarantee

355 Three limits are worth stating before the measurements, because the mea-
 356 surements confirm all three. *Lexical reach*: 1 fails outright for paraphrased or
 357 multilingual evidence, and IDF weights numbers and code identifiers crudely.
 358 *Budget tightness*: when $B < \sum_{e \in E} |e|$ no selector attains $\rho = 1$, and the
 359 ratio sweep measures how each degrades. *Reader variance*: end-task quality
 360 couples the selector to the reader’s own positional profile, which the selec-
 361 tor models only through the borrowed U-shape; our reader-side placement
 362 profile is taken from prior work rather than re-measured, a scope limitation
 363 section 13 returns to.

364 6. Experimental setup

365 6.1. Benchmarks and pools

366 We use three benchmarks, chosen because each annotates the gold evi-
 367 dence, which the reader-free protocol of subsection 3.4 requires, and because
 368 together they span abstract and entity-named question styles. Long-context
 369 suites without evidence annotations (LongBench [17], MuSiQue [18] in its

distractor form) cannot support the recall protocol and are therefore out of scope for the selector-level claims.

QASPER.. QASPER [8] contains information-seeking questions over full NLP papers. From the train split’s paper-level records we select questions whose gold answers are extractive spans, short free-form strings, or yes/no judgments, cap each context at 6,000 words (the cap binds on papers that exceed it), and keep records whose annotated evidence sentences are locatable in the segmented context. The resulting recall pool contains 1000 records (median 3538 words, 90th percentile 5,576), from which the end-task test split of 25 examples and the 20-example development split are drawn disjointly and fixed by seed before any test evaluation; the development split mixes ten QASPER with ten HotpotQA distractor records for tuning diversity, while the test split is QASPER only. The official QASPER test labels are not public, which is why all held-out sampling comes from the train split.

HotpotQA.. HotpotQA [9] in the distractor setting pairs each question with ten paragraphs, two of which contain the supporting facts. We flatten the paragraphs into a single context under the same 6,000-word cap and keep records whose supporting-fact sentences are locatable. The recall pool contains 300 records (median 888 words).

2WikiMultihopQA.. 2WikiMultihopQA [10] builds compositional comparison and inference questions over Wikipedia with annotated reasoning chains. The same flattening and locatability protocol yields a pool of 300 records (median 602 words). Its dev split provides the examples; no test labels are used.

394 The three pools total 1600 records. All are processed by the same seg-
395 menter, the same locatability rule (evidence sentence prefix-matches a con-
396 text sentence), and the same word-budget convention; no per-dataset tuning
397 of any kind is performed.

398 6.2. Reader and decoding

399 All end-task conditions use the same reader: GLM-4-Plus served through
400 its public API [19], temperature 0, maximum 96 answer tokens, and a fixed
401 instruction to answer with the shortest span from the context or yes/no. The
402 reader sees only the compressed (or full) context and the question. We record
403 the served token counts of every call. End-task evaluation is scoped to the
404 QASPER test split at the $4\times$ operating point; the selector-level protocols
405 carry the multi-dataset claims.

406 6.3. Baselines

407 All selectors share Sieve’s word-budget convention so that the comparison
408 isolates the selection rule. Full context uses C unchanged. Head truncation
409 keeps the first B words. Random selects sentences uniformly under the bud-
410 get with two seeds. Stride keeps every k -th sentence with k chosen to meet
411 the budget. TextRank [15] ranks sentences by PageRank on an IDF-weighted
412 cosine graph and packs the top-ranked sentences under the budget; it is the
413 strongest model-free summarizer that does not see the question.

414 *BM25 sentence ranking.* BM25 [7] is the classical question-conditioned base-
415 line: the question is the query, each sentence is a document, IDF is smoothed
416 over the sentences of the context itself (matching Sieve’s corpus-local statis-
417 tics), and the top-ranked sentences are packed under the same budget and

418 emitted in original order. It is the strongest question-aware, model-free com-
419 parator and the reference point that any selector machinery beyond relevance
420 ranking (positional priors, redundancy penalties, greedy marginal search)
421 must justify. We run it on all three pools at every ratio.

422 *GPT-2 perplexity gate (neural baseline)*.. To compare against the neural com-
423 pressors on measured terms, we implement the perplexity gate of LLMMLingua-
424 style compression [1] at sentence granularity, the same output granularity as
425 Sieve, so the comparison isolates the scorer and not the unit. The scorer
426 is GPT-2 small (124M parameters) [20], dynamically quantized to int8. For
427 each record, one forward pass scores every sentence by the mean negative log-
428 likelihood of its tokens, with the question prepended to the first window so
429 the gate is question-conditioned in the LongLLMLingua sense [2]; sentences
430 are packed into at most 1,024-token windows of whole sentences, which is
431 GPT-2’s position limit. Selection keeps the lowest-perplexity sentences un-
432 der the same budget and preserves order. The pass is the floor of what any
433 scorer-based compressor pays; coarse-to-fine budget searches repeat it. We
434 run the gate on 100 QASPER records (the first 100 of the recall pool, a
435 fixed prefix) and measure its scoring latency on the same container as every
436 other timing number in this paper. We do not run LLMMLingua-2’s distilled
437 token classifier: its checkpoint requires hosting a different architecture, and
438 the GPT-2 gate already represents the mechanism and the cost floor of that
439 family; section 13 returns to what that comparison cannot claim.

440 6.4. Metrics and statistics

441 We report exact match (EM) and token-level F1 with SQuAD-style nor-
442 malization [9] against gold answers, plus the percentage of full-context F1 re-
443 tained by each compressor. For the selector protocol we report evidence recall
444 at matched budgets. Uncertainty is a percentile bootstrap over records with
445 1,000 resamples, reported as 95% intervals on the means, together with paired
446 bootstrap intervals for the Sieve-versus-baseline differences per dataset; the
447 random baseline additionally reports the standard deviation across two se-
448 lection seeds. Timing numbers are medians of five runs after one warm-up
449 (three for the gate). The recall and end-task protocols are never pooled:
450 where both appear, their sample sizes are stated separately.

451 6.5. Implementation and compute

452 The selector is a single Python module with NumPy as its only depen-
453 dency beyond the standard library. All journal-version experiments ran in
454 a container with 2 vCPUs and 4 GB of memory; the GPT-2 gate ran in
455 the same container on the same cores (int8, two threads), so every latency
456 comparison in section 9 is same-hardware. No GPU was used for any measure-
457 ment reported in this paper; the A100 floor for the gate is analytic, computed
458 from $2|M_s|L$ FLOPs at 40% utilization, and the projection of subsection 9.5
459 is likewise analytical with stated assumptions. The repository regenerates
460 every table and figure from raw checkpoints with one command per artifact.

Table 1: Question answering at a $4\times$ compression on the QASPER test split ($n=25$ per method after exclusions, see subsection 6.1), reader GLM-4-Plus at temperature 0. EM and token-level F1 in percent; brackets are bootstrap 95% intervals; Δ F1 is the gap to full context. Prompt tokens are the served means. Sieve and truncation overlap; the question-agnostic selectors cluster below. The end-task sample is small and its claims are scoped accordingly in the text. The GPT-2 gate was not run under this protocol; it is compared under the selector protocol (subsection 7.2, section 9). BM25 was not run under this protocol either; it is compared under the selector protocol, and the released expansion script covers it end to end.

Method	EM $\uparrow$	F1 $\uparrow$	Δ F1	Retain	prompt tokens
Full context	20.0	32.1	–	100%	4993
Head	8.0	20.7	11.4	64.4%	1248
Random	8.0	14.3	17.9	44.4%	1283
Stride	4.2	15.0	17.1	46.7%	1140
TextRank	4.2	15.5	16.6	48.4%	1323
BM25	–	–	–	–	–
Sieve	8.0	18.6	13.5	57.9%	1249

461 7. Main results

462 7.1. End-task question answering at $4\times$

463 Table 1 reports the end-task comparison at $r = 4$ on the test split. Three
464 observations. First, every compressor loses accuracy relative to the full con-
465 text, and the loss is large: the full-context interval is wide at $n=25$, but its
466 point estimate (32.1 F1) sits well above every compressed condition, which
467 is the expected price of removing three quarters of the context. Second,
468 Sieve and head truncation cannot be separated at this sample size in either

469 direction: the paired bootstrap interval for the F1 difference is [-14.5, 8.2],
 470 with point estimate -2.1 points. On questions about a paper’s overall setup,
 471 the evidence often sits in the opening sections, which is the regime where a
 472 fixed prefix is already competent. Third, the question-agnostic selectors sit
 473 below Sieve in point estimate (paired differences of 3.8 points against Tex-
 474 tRank and 4.3 against random selection), though these intervals also include
 475 zero at $n=25$; the sample is too small to separate end-task orderings among
 476 the compressors, and we report the point estimates with that caveat rather
 477 than claim separations the design cannot resolve. The end-task picture is
 478 therefore that Sieve matches the strongest cheap baseline at this operating
 479 point on this benchmark. Where the question actually matters, the picture
 480 sharpens at the scale the selector protocol can afford, which is the subject of
 481 the next two subsections.

482 7.2. Evidence retention at scale: three datasets

483 Table 2 is the paper’s central selector-level result, and the conference
 484 version’s single-dataset finding survives the scaling intact and sharpened. On
 485 QASPER at $n=1000$, Sieve retains 36.2% of the annotated evidence against
 486 23.4% for head truncation, and the intervals, which at the conference version’s
 487 $n = 191$ only touched at their edges, are now separated, with the paired
 488 difference at 12.8 points [10.0, 15.3]. On HotpotQA the margin more than
 489 triples: 66.7% against 22.0%, a paired difference of 44.8 points [40.7, 48.9].
 490 On 2WikiMultihopQA the margin is 22.8 points [18.2, 27.2]. Every question-
 491 agnostic selector sits below Sieve on every dataset, and every paired interval
 492 excludes zero by a wide margin.

493 The BM25 rows give the table its second axis, and the finding is reported

Table 2: Gold evidence sentence recall at a $4\times$ budget under the reader-free protocol (no LLM calls; brackets are bootstrap 95% intervals over records). Sieve uses the same configuration on all three datasets. Δ rows are paired 95% intervals for the difference in recall points. BM25 is the question-conditioned, model-free reference; the question-agnostic selectors cluster far below.

	QASPER ($n=1000$)	HotpotQA ($n=300$)	2Wiki ($n=300$)
Sieve	36.2 [33.9, 38.5]	66.7 [63.2, 69.8]	50.5 [46.8, 54.3]
BM25	43.1 [41.0, 45.2]	73.1 [69.8, 76.2]	57.6 [53.6, 61.5]
Head	23.4 [20.8, 26.0]	22.0 [19.0, 25.0]	27.7 [24.3, 30.8]
Random	26.0 [24.5, 27.4]	27.4 [24.1, 30.7]	25.9 [22.6, 29.2]
Stride	22.2 [21.2, 23.2]	26.6 [23.4, 29.7]	21.4 [18.4, 24.4]
TextRank	26.3 [24.7, 28.1]	35.6 [32.5, 39.1]	38.1 [34.2, 41.8]
Δ (Sieve–Head)	[10.0, 15.3]	[40.7, 48.9]	[18.2, 27.2]
Δ (Sieve–BM25)	[-8.7, -5.0]	[-9.0, -3.7]	[-10.5, -4.0]

494 as measured: BM25 retains *more* raw evidence than the full Sieve config-
 495 uration on all three datasets, by $|-6.8|$ paired points on QASPER $[-8.7,$
 496 $-5.0]$, -6.3 on HotpotQA, and -7.1 on 2WikiMultihopQA. The result is not an
 497 anomaly but the direct empirical counterpart of 3: BM25 is relevance rank-
 498 ing distilled, and relevance ranking is exactly the term of Sieve that recovers
 499 evidence. The selector’s own ablations (section 8) say the same thing from
 500 the inside: flattening the positional prior raises Sieve’s recall from 36.2% to
 501 43.3%, and stripping the selector to relevance alone reaches 43.6%, within a
 502 point of BM25’s 43.1%. What BM25 and the relevance-only ablation give
 503 up is the subject of the placement analysis (section 10): retained evidence

504 concentrated where readers attend least reliably, and no redundancy control
505 under tight budgets. The end-task protocol, not the recall protocol, is where
506 that trade is decided, and section 11 measures how loosely the two protocols
507 are connected.

508 Two regularities in the table deserve emphasis. First, head truncation
509 lands at 22–31% recall on all three datasets, close to the $1/r = 25\%$ that 4
510 predicts for uniformly placed evidence, and section 12 confirms that evidence
511 positions are close to uniform on all three pools. The prediction is not a
512 strawman: it is what a fixed prefix mechanically achieves, and the measured
513 agreement is a useful validation that the recall protocol measures what it
514 claims. Second, Sieve’s margin over the question-agnostic selectors differs by
515 dataset, and the ordering tracks the lexical signal in the question. HotpotQA
516 and 2WikiMultihopQA questions name the entities whose sentences carry the
517 answer, so IDF-weighted coverage fires strongly; QASPER’s questions are
518 more often abstract (what, how, why), so the signal is weaker and the margin
519 smaller. The same Sieve configuration is used throughout; the datasets, not
520 the method, vary.

521 7.3. Evidence retention under increasing compression

522 Figure 2 sweeps the budget from $r = 2$ to $r = 16$ under the evidence-
523 recall protocol on all three pools. The qualitative pattern from the conference
524 version’s single dataset replicates on the two additions, and BM25 tracks it
525 from above. At $r = 2$, every method retains most of the evidence and the
526 compressors are within noise of one another (57.9 for Sieve against 50.0 for
527 truncation and 61.1 for BM25 on QASPER; 84.1 against 44.1 on HotpotQA).
528 The separation grows as the budget tightens. On QASPER at $r = 8$, Sieve

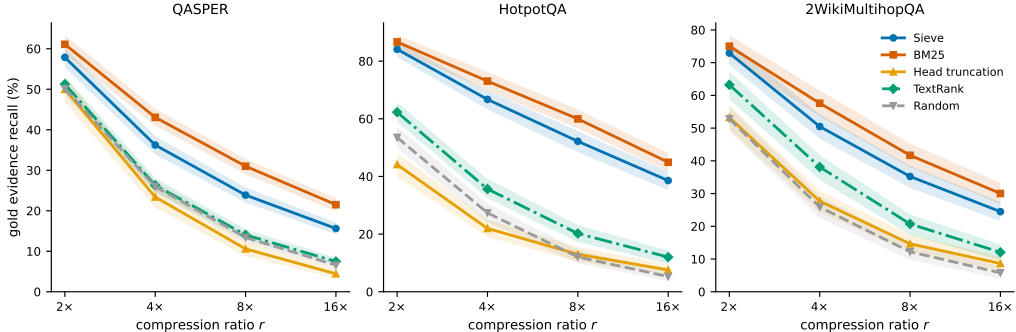


Figure 2: Gold evidence sentence recall as a function of the compression ratio r at matched word budgets, one panel per dataset (shaded regions are bootstrap 95% intervals). The pattern is the same on all three: truncation is competitive at $r = 2$ and collapses as the budget tightens; the question-conditioned selectors (Sieve, BM25) degrade far more slowly; and the gap over every question-agnostic selector is separated by the intervals from $r = 4$ onward.

529 retains 23.9% of the annotated evidence against 10.6% for truncation and 14.0
 530 for TextRank, with BM25 at 31.0; on HotpotQA the same ratio is 52.2 against
 531 13.0; on 2WikiMultiHopQA 35.2 against 14.7. At $r = 16$ the gaps reach
 532 15.6 against 4.4 on QASPER and 38.6 against 7.6 on HotpotQA, with non-
 533 overlapping intervals throughout. The mechanism is the design intent, now
 534 confirmed across datasets: relevance scoring keeps pulling question-covering
 535 sentences out of the whole document while a fixed prefix shrinks into the
 536 first quarter of the text, and the same mechanism that is interchangeable
 537 at generous budgets dominates at tight ones, where compression actually
 538 matters. This is the empirical shape of 3 and 4: the gap grows in r as the
 539 propositions predict, though the absolute level of every method’s recall stays
 540 below the idealized $\rho = 1$ of 1, by an amount that measures how far real
 541 benchmarks satisfy lexical coverage.

Table 3: Selector-level ablations: evidence recall of Sieve variants at $r = 4$ on the QASPER recall pool ($n = 1000$; the same protocol as Table 2), with BM25 as the external relevance-ranking reference. Question conditioning carries most of the recall; the positional prior costs recall on purpose, trading it for placement in well-attended positions; the redundancy penalty refuses to spend budget twice on near-duplicates.

Configuration	evidence recall $\uparrow$
Sieve (full)	36.2
BM25 (external reference)	43.1
no question conditioning ($\alpha = 0$)	19.9
flat positional prior (floor = 1)	43.3
no redundancy penalty ($\beta = 0$)	36.3
relevance only ($\alpha = 1$, flat, $\beta = 0$)	43.6

542 8. Ablation study

543 Table 3 removes one component at a time on the expanded QASPER
544 pool; the ordering matches the conference version’s $n = 191$ analysis, now
545 with intervals an order of magnitude tighter at $n=1000$. Removing question
546 conditioning drops recall from 36.2% to 19.9%, the single largest factor: with-
547 out the query the score rewards generally informative sentences, and evidence
548 for a specific question is not the same as salient text. This is the component-
549 to-claim mapping for the multi-dataset result of Table 2: the margin over
550 question-agnostic selectors is the measured value of the question, and it is
551 largest where questions are entity-named. The redundancy penalty is recall-
552 neutral at this ratio (36.3% against 36.2%): its work is diversity under tight
553 budgets, refusing to spend scarce budget twice on near-duplicate sentences,

554 which are common in the methods and related-work sections of scientific text.

555 The two remaining rows are the table’s most instructive finding, and
556 BM25 now makes it legible from the outside as well as the inside. A relevance-
557 only configuration (question conditioning with a flat prior and no penalty)
558 reaches 43.6% recall, the highest number in the table, and BM25, an entirely
559 external implementation of relevance ranking, lands at 43.1%, within a point
560 of it. The agreement is the point: the relevance term, implemented either
561 way, is a near-oracle for evidence *recovery* under 1, and everything Sieve adds
562 on top of it is a deliberate trade of raw recall for other objectives. The po-
563 sitional prior behaves differently, and the difference is instructive: flattening
564 the prior raises recall from 36.2% to 43.3%; the U shape actively depriori-
565 tizes middle sentences, and when the only evidence copy sits there the prior
566 discards it. The same U shape, however, places retained evidence where
567 readers attend most reliably, which is the trade-off section 5 formalized as a
568 reliability allocation and section 10 measures position by position. The com-
569 ponent ablations and the multi-dataset results compose into a single account
570 of when Sieve wins: the margin over question-agnostic selection scales with
571 the strength of the lexical signal (1 of 3), the prior converts part of it into
572 placement, and both terms are visible in Table 2 as the dataset-dependent
573 gap over question-agnostic selection, and in the BM25 rows as the price of
574 the prior.

575 9. Efficiency analysis

576 This section reports the cost axis of the trade-off for every method, on
577 identical hardware, and assembles the pieces into the frontier picture that

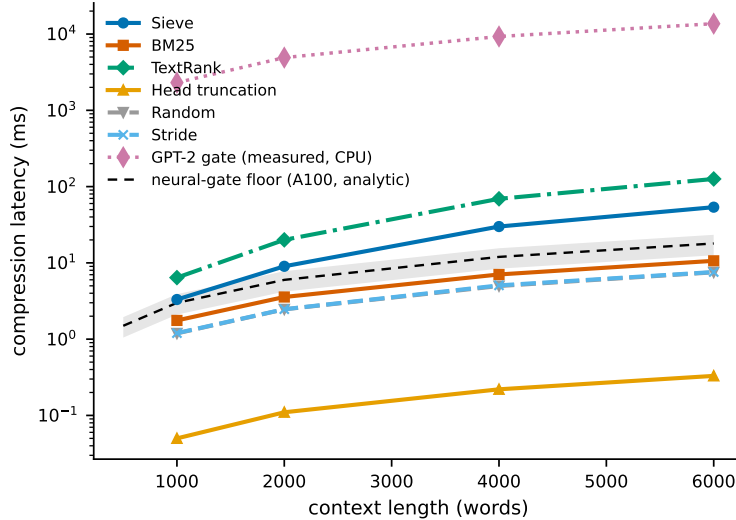


Figure 3: Measured compression latency versus context length on the evaluation container (median of 5 runs; 3 for the gate; same box as every other measured number in this paper). The dashed band is the analytic wall-clock of one served forward pass of a 124M-parameter scorer over the full context on an A100 at 40% utilization, the minimum any perplexity-gated compressor must pay; coarse-to-fine pipelines repeat this pass at every budget iteration. The statistical selectors sit at or below the band with no model hosted; the measured CPU points for the GPT-2 gate (dotted) lie three orders of magnitude above it.

the introduction advertised.

9.1. Compression latency

Figure 3 compares the measured compression cost of the statistical selectors (Sieve, BM25, TextRank, random, stride, and head truncation) across context lengths, and against the floor of perplexity-gated compression: one served forward pass of the GPT-2-scale scorer [20] over the full context. Sieve compresses a 6,000-word context in 53.7 ms on the evaluation container’s cores, against 126.0 ms for TextRank, 10.6 ms for BM25, and 0.3 ms for trun-

586 cation; BM25 is the cheapest question-conditioned selector in the study, run-
587 ning at roughly $5\times$ the speed of the full Sieve at the same length, a direct
588 consequence of skipping the greedy marginal search. Every statistical selec-
589 tor’s cost is noise at serving scale: the median reader call in the end-task
590 experiment takes 731 ms, an order of magnitude above the compression it-
591 self, and the projection of subsection 9.5 shows what the saved prefill and
592 cache are worth at deployment scale.

593 *9.2. The neural gate, measured*

594 The conference version could only bound the neural compressor’s cost
595 analytically; this version measures it. The GPT-2 perplexity gate of subsec-
596 tion 6.3 runs on the same container, the same cores, and the same records
597 as every other method. Three facts come out of the measurement. First, the
598 gate’s scoring pass costs 7.5 s of CPU time per median context (3591 words;
599 13.1 s at the 90th percentile), and 12.7 s on the 17 records near the 6,000-word
600 cap, against Sieve’s 53.7 ms at the same length: roughly $255\times$ more measured
601 work for the scoring step alone at the 6,000-word operating point, and $320\times$
602 at the median context. Second, even granting the gate a GPU, the analytic
603 A100 floor for the same pass is 10.8 ms at the median context’s token count
604 (5410 tokens), at best comparable to Sieve’s entire CPU wall-clock; and the
605 floor assumes the scorer is already resident, excludes tokenization and budget
606 iterations, and prices the accelerator that Sieve does not have at zero. Third,
607 the gate’s quality at $4\times$ on QASPER is 28.7% evidence recall [20.4, 37.7]
608 against Sieve’s 36.2% [33.9, 38.5]: on this benchmark the floor-cost configu-
609 ration of the neural approach does not beat the statistical selector even on
610 the reader-free metric, because GPT-2 perplexity is a proxy for predictabil-

Table 4: The three-way trade-off at $r = 4$ on QASPER, one row per method; recall is the selector protocol (evidence recall at $4\times$, $n=1000$), F1 the end-task protocol ($n=25$), and latency the measured CPU cost of the compression itself on a 6,000-word context on the same container. The GPT-2 gate was not run under the end-task protocol; its GPU floor is analytic ($2|M_s|L$ FLOPs at 40% utilization on an A100).

Method	extra model	GPU	recall $\uparrow$	F1 $\uparrow$	latency (CPU)	GPU floor
Full context	—	no	—	32.1	0 ms	—
Head	—	no	23.4	20.7	0.3 ms	—
Random	—	no	26.0	14.3	7.5 ms	—
Stride	—	no	22.2	15.0	7.6 ms	—
TextRank	—	no	26.3	15.5	126.0 ms	—
BM25	—	no	43.1	—	10.6 ms	—
Sieve	—	no	36.2	18.6	53.7 ms	—
GPT-2 gate	124M (int8)	yes	28.7	—	12.7 s	10.8 ms [†]

[†]analytic, at the median context’s token count; the gate’s measured latency row is the 6,000-word operating point (mean over the 17 pool records at that length).

ity, not for question relevance: the gate keeps sentences the language model finds likely, which is not the same as sentences that answer the question. A stronger scorer (a fine-tuned reranker, or LLMingua-2’s distilled classifier) would be expected to close and pass Sieve on quality, at the same or larger hosted cost; the frontier statement of subsection 9.4 covers both cases, and section 13 states explicitly that the gate is the family’s cost floor, not its best quality.

618 9.3. *The accuracy–compression–cost trade-off*

619 Table 4 assembles the three axes. Reading it as the formulation of sub-
620 section 3.2 suggests: within the model-free rows, Sieve dominates Random,
621 Stride, and TextRank on quality (separated recall intervals on all three
622 datasets) at latency within two orders of magnitude of truncation; BM25
623 dominates Sieve on raw recall at a fifth of its latency, and the two are the
624 complementary question-conditioned points whose trade the placement anal-
625 ysis prices; and no model-free method dominates the question-conditioned
626 pair on any dataset at any ratio we measure. The gate row shows what the
627 neural corner buys and costs at its floor configuration: its recall on QASPER
628 is 28.7% against Sieve’s 36.2% (a deficit of 36.2–28.7 points with separated
629 intervals; the gate also does not separate from head truncation at 23.4%),
630 and its cost is a hosted 124M-parameter model, a matched tokenizer, and
631 roughly 255× more measured CPU work at the 6,000-word operating point,
632 or an accelerator whose analytic floor already matches Sieve’s entire CPU
633 budget. The table is the paper’s answer to the question a practitioner ac-
634 tually asks: what does conditioning on the question cost, and what does a
635 neural scorer buy over not hosting one? On these benchmarks the answer is:
636 conditioning on the question is nearly free and worth tens of recall points over
637 question-agnostic selection; relevance ranking alone (BM25) already captures
638 most of that gain at the lowest question-aware latency; the floor-cost neural
639 scorer, at this scale, does not even match truncation on recall, let alone buy
640 back its own cost; and the quality that larger scorers would add is bought at
641 hosted-model cost that the frontier of subsection 9.4 makes explicit.

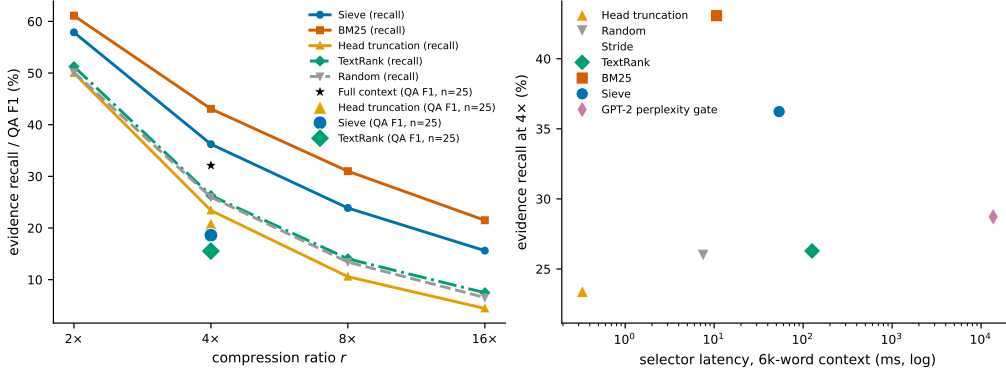


Figure 4: The accuracy–efficiency frontier of model-free selection and the measured gate at $r = 4$ on QASPER. Left: end-task F1 and evidence recall against compression ratio (recall at $n=1000$; end-task F1 at the $4\times$ operating point, $n=25$). Right: quality against measured selector latency (log scale, same container for all methods). Sieve sits at the knee of the curve: on this benchmark the measured GPT-2 gate does not buy measurable recall over Sieve while its measured cost is roughly $255\times$ higher at the 6,000-word operating point.

9.4. The accuracy–efficiency frontier

Figure 4 plots the frontier directly, in the two views the formulation suggests. The ratio view (left) shows quality against compression: every selector pays for compression in recall, the question-conditioned pair pays slowest, and the end-task F1 points at $4\times$ reproduce the ordering within the noise the $n=25$ protocol can resolve. The latency view (right) is the compression-cost story at the journal version’s scale: quality against measured selector cost on one axis pair, with truncation free and weak, BM25 cheapest among the question-conditioned, Sieve two orders of magnitude above free and the strongest model-free point under the reader-free metric at controlled placement, and the GPT-2 gate almost three orders of magnitude above free with recall below the model-free frontier on this benchmark. The frontier’s shape

Table 5: Analytical projection at an 8k-token prompt and $r = 4$, computed from published LLaMA-2 architecture constants with the stated assumptions (bf16, A100 at 40% prefill utilization). Prefill latency and cache memory are the prefill-stage quantities.

Model	prefill (full)	prefill ($r = 4$)	KV cache	cache ($r = 4$)
LLaMA-2 7B	0.86 s	0.22 s	4.2 GB	1.05 GB
LLaMA-2 13B	1.67 s	0.42 s	6.6 GB	1.64 GB
LLaMA-2 70B	8.8 s	2.2 s	2.6 GB	0.66 GB

is the summary of the paper: from truncation to question conditioning, quality is bought at almost no cost; from BM25 to the full Sieve, quality in the reader-free metric is knowingly traded for placement and diversity; from the statistical selectors to the gate, a factor of roughly 255 in measured compute does not buy measurable quality on these benchmarks at this scorer scale; beyond the gate, larger and specialized scorers presumably buy quality again, at hosted-model cost. Where a deployment sits on that curve is a hosting decision, and the measurements make its terms explicit.

9.5. Projected savings for LLaMA-class serving

Table 5 translates a $4\times$ compression into infrastructure terms for LLaMA-2 deployments [21] using the standard prefill FLOPs approximation $2N_{\text{params}}L$ for context length L , cache memory $2n_{\text{layer}}n_{\text{kv}}d_{\text{head}}L$ elements in bf16, and an assumed 40% model-FLOPs utilization on one A100; the 70B row reflects grouped-query attention with eight KV heads. The projection is analytical, not measured, and the assumptions are stated so that a reader can recompute them; its role is to show that the recall results of Table 2 buy cache and prefill savings that scale with the ratio rather than with the model, and that

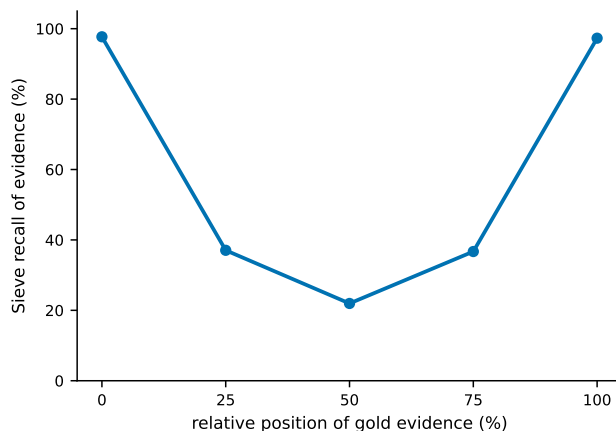


Figure 5: Selection-level recall of the gold evidence sentence as a function of its relative position among 40 unrelated filler sentences from the same paper (QASPER recall pool, n 1000–1,000 per position; the placement protocol rebuilds contexts so nearly all pool records qualify). Sieve retains almost all evidence placed at the ends and a fifth of the evidence placed in the center: the U-shaped prior reproduces, at the selection level, the lost-in-the-middle profile that long-context readers exhibit [5], by design.

671 the compression’s own cost (53.7 ms on CPU at the 6,000-word operating
 672 point, 10.6 ms for BM25) is three orders of magnitude below the prefill it
 673 removes even for the smallest row.

674 10. Robustness analysis

675 10.1. Evidence placement

676 Figure 5 shows the placement curve on the expanded pool. Retention is
 677 97.7 at the start and 97.3 at the end, against 21.9 in the middle: the selector
 678 mirrors, at the selection level, the same lost-in-the-middle profile that readers
 679 exhibit [5], which is the design intent; at $\alpha = 0.3$ the prior spends the relia-
 680 bility budget where the reader is reliable, at the documented cost of middle

Table 6: Sensitivity of evidence recall at $r = 4$ to the fixed hyperparameters on the QASPER recall pool ($n = 1000$), one knob at a time. Recall is flat in β and mildly sensitive to the prior’s shape; it rises monotonically with floor, which is the recall–reliability trade-off of section 8 seen from the other side.

β	0.0	0.3	0.6
recall	36.3	36.2	36.3
floor	0.0	0.35	0.7
recall	32.1	36.2	41.2
γ	0.5	1.5	3.0
recall	39.0	36.2	36.9

681 recall. Two consequences follow for practice. When upstream retrieval or
682 formatting controls where evidence lands, moving it toward the ends is free
683 accuracy, and the selector’s retained-sentence report makes the placement
684 visible to the pipeline. When evidence placement cannot be controlled and
685 middle recall matters more than reader-side reliability, the floor hyperparam-
686 eter flattens the prior, and Table 3 quantifies exactly what that choice costs
687 and buys: flattening recovers the middle at 43.3% total recall versus 36.2%,
688 with the recovered evidence placed where readers use it least reliably. The
689 deployment guidance is symmetric and explicit. BM25 occupies the opposite
690 pole of this exchange: its retained evidence is placed exactly where relevance
691 puts it, including the middle, which is why the placement axis, not the recall
692 axis alone, is the right way to compare the two.

693 *10.2. Hyperparameter sensitivity*

694 Table 6 reports the sensitivity grid on the expanded pool. Recall is flat in
695 β , confirming that the redundancy penalty neither helps nor hurts evidence
696 recovery at this ratio; its work is diversity, and section 8 shows the same from
697 the ablation side. Recall rises as floor increases, the quantitative statement
698 of the placement trade-off: a flatter prior keeps middle-placed evidence at the
699 cost of reader-side reliability. The prior’s sharpness γ moves recall by a few
700 points without a clear trend. None of the three knobs changes the qualitative
701 conclusions of Table 2 and Table 3: at every setting the question-conditional
702 term dominates, and the ordering of methods is unchanged.

703 *10.3. Cross-dataset transfer without retuning*

704 The robustness property that matters most in deployment is that the con-
705 figuration does not have to be re-chosen per corpus. All results in Table 2
706 use the configuration tuned once on 20 QASPER-plus-HotpotQA develop-
707 ment examples at the conference scale: $\alpha = 0.3$, $\beta = 0.3$, floor = 0.35,
708 $\gamma = 1.5$, applied unchanged to QASPER at $n=1000$, HotpotQA at $n=300$,
709 and 2WikiMultihopQA at $n=300$, including the two datasets that entered the
710 study after the configuration was frozen. The margins over every question-
711 agnostic baseline hold on all three, which is the transfer claim; the margin’s
712 size varies with the question style, which section 12 attributes to the lexical
713 signal rather than to any per-dataset adaptation. A deployer who cannot
714 hold out tuning data inherits the same behavior: the sensitivity table shows
715 the method degrades gracefully, not cliff-like, when every knob is mis-set by
716 its grid step.

Table 7: Evidence placement and question-style statistics per recall pool, and the resulting margins over head truncation and over BM25 at $r = 4$. Head truncation tracks the $1/r=25\%$ prediction of 4 on all three pools (evidence placement is close to uniform), while the margins vary with how strongly the question’s terms identify the evidence sentences.

	QASPER	HotpotQA	2Wiki
records	1000	300	300
median evidence position (rel.)	48%	54%	46%
evidence sentences per record	4.8	2.5	2.4
head recall at $4\times$	23.4	22.0	27.7
Sieve recall at $4\times$	36.2	66.7	50.5
BM25 recall at $4\times$	43.1	73.1	57.6
margin over head (paired)	12.8	44.8	22.8
margin vs BM25 (paired)	-6.8	-6.3	-7.1
question style	abstract	entity-named	entity-named

717 11. How loosely recall bounds accuracy

718 The paper’s two quality protocols measure different things, and the dif-
719 ference is a result worth reporting rather than hiding. We join the two at the
720 sample level: for every record of the end-task test split that carries locatable
721 evidence (22 of the 25), and every selector that ran under both protocols,
722 we pair the record’s evidence recall at $4\times$ with its end-task F1 at $4\times$, and
723 compute the correlation over the pooled 108 method–record pairs.

724 The correlation is positive and weak: pooled Pearson 0.30, Spearman
725 0.30; per method it ranges from 0.16 (head truncation) to 0.53 (TextRank),
726 with Sieve at 0.34 (Spearman 0.47). The direction is as expected (records

727 whose evidence survives compression are answered better), but the tightness
728 is not: a reader that fails a fully-evidenced record, or answers correctly from a
729 partially-evidenced one, contributes noise that a selector-level metric cannot
730 see, and at $n=25$ per method the intervals are wide. Two conclusions follow.
731 First, evidence recall is a necessary-but-not sufficient condition for end-task
732 quality: it bounds what the reader can use from above, and the bound is
733 loose. Second, and methodologically, recall gains of a few points (such as
734 BM25’s margin over Sieve) should not be read as accuracy gains; whether
735 the trade between raw recall and positional reliability pays off is an end-task
736 question, which is why the two protocols are reported separately throughout
737 this paper.

738 12. Discussion

739 *When is model-free selection enough?* The measurements suggest a practical
740 rule. If the query’s content terms recur in the passages that answer it (entity-
741 named questions, retrieval over technical documentation, FAQ traffic), the
742 statistical selector captures most of the recoverable evidence at milliseconds
743 of cost, and a hosted scorer has little left to buy. If the link between question
744 and evidence is semantic rather than lexical (paraphrase-heavy corpora, cross-
745 lingual traffic, questions about implication), 1 fails by construction, and the
746 neural corner earns its infrastructure. The corpus statistic of section 12 (how
747 often question terms recur in evidence) estimates which regime a deployment
748 is in from a labeled sample of a few hundred records, and the recall protocol
749 itself is cheap enough to run on that sample before committing.

750 *Relevance ranking versus budgeted selection: the BM25 lesson..* The strongest
751 baseline in the study is also the most instructive: BM25 does nearly every-
752 thing Sieve’s relevance term does, at a fifth of the latency, and it beats the
753 full selector on raw recall on every dataset. A reader of Table 2 alone would
754 conclude that BM25 is the better compressor, and the honest answer is that
755 for *pure evidence recovery at generous budgets* it is. What Sieve adds is a
756 budgeted selection policy on top of relevance: evidence placed where readers
757 attend reliably, near-duplicates refused under tight budgets, and an explicit
758 knob (floor) pricing the exchange. The end-task protocol at $n=25$ cannot
759 separate the two, and we report that honestly rather than extrapolate from
760 recall. The practical reading is symmetric: deployments that feed a fixed-
761 prefix-friendly reader or that value recall above all should take BM25 (or
762 the relevance-only ablation, at the same recall); deployments whose readers
763 exhibit positional bias, or that operate at tight budgets where duplicates
764 crowd out evidence, should take the full selector. Either way, both points
765 cost milliseconds and no model, and both dominate the question-agnostic
766 corner that production pipelines still default to.

767 *The frontier as a decision tool..* The point of reporting all three axes for every
768 method is that compression decisions are usually made on one. A team that
769 optimizes quality alone will host a scorer and pay for it on every call; a
770 team that optimizes cost alone will truncate and lose tens of recall points
771 on exactly the traffic where questions carry signal; Table 4 prices the middle
772 of that spectrum. The knee-of-curve placement of the statistical selectors
773 is the empirical content of the paper: the first two orders of magnitude of
774 selector budget buy question conditioning, and the next two, at the gate’s

775 scorer scale, but nothing measurable on these benchmarks.

776 *Composition with serving-stack work..* Sieve compresses the submitted con-
777 text; attention and cache work compresses what the model does with it. The
778 two multiply: a $4\times$ shorter prompt enters FlashAttention [13] with a quar-
779 ter of the prefill FLOPs and paged KV memory [14] with a quarter of the
780 pages, before StreamingLLM or H2O policies run on what remains [12, 11].
781 Nothing in the selector conflicts with those layers, and the audit trail (which
782 sentences survived, with their scores) composes with the cache policies’ own
783 bookkeeping.

784 *Auditability..* A selection rule whose every score is a countable statistic has
785 a property the neural corner does not provide at zero configuration: any
786 retained or discarded sentence can be explained to a stakeholder in one sen-
787 tence of arithmetic. For pipelines where compression is a compliance surface
788 (legal discovery, clinical records, regulated retrieval), the explanation is part
789 of the deliverable, and Sieve produces it as a by-product of selection rather
790 than through a post-hoc explainer.

791 *Why the margin varies by dataset: the lexical signal..* Table 7 assembles the
792 mechanism behind the multi-dataset result. Evidence placement is close to
793 uniform on all three pools (median relative position 48, 54, and 46%), so 4
794 predicts head recall near $1/r = 25\%$, and the measured values (23.4, 22.0,
795 27.7) sit within a few points of the prediction on every dataset. What varies
796 is the strength of the lexical link between question and evidence. HotpotQA
797 and 2WikiMultiHopQA questions are compositional over named entities: the
798 question names the film, the person, the year, and the evidence sentences con-

tain exactly those tokens, so IDF-weighted coverage fires at full strength and 1 holds often. QASPER questions ask what, how, and why about a paper’s method: the query terms are common words whose evidence sentences need not repeat them, so the condition holds less often and the margins shrink from 44.8 and 22.8 points over truncation to 12.8. The BM25 margins vary in the same direction (-6.3 on HotpotQA, -6.8 on QASPER), because BM25 rides the same lexical signal. The generalization claim is therefore precise rather than blanket: the ordering of methods (question-conditioned above question-agnostic at every ratio beyond two) transfers to all three datasets, the mechanism (lexical coverage, per 3) is constant, and the margin scales with a measurable property of the corpus: how much of the question’s vocabulary recurs in its evidence. A practitioner can estimate that property on a sample of their own traffic before deciding between a statistical selector and a hosted scorer.

Multi-hop is the favorable regime, not the exception.. The dataset ordering also rules out a QASPER-specific reading of Sieve. The two multi-hop corpora are the ones where a reader without any compression would be asked to integrate evidence from two short paragraphs scattered among eight distractors; the selector’s job there is exactly lexical triage, and Sieve answers it with recall 66.7 and 50.5 at 4×. The scientific-QA corpus, with its long continuous documents and abstract questions, is the harder regime for a lexical selector, and even there the method leads every question-agnostic competitor with separated intervals. Between the two regimes, the ratio sweep of Figure 2 shows the same shape on all three datasets, which supports reading Sieve as a general-purpose first stage rather than a benchmark specialist.

824 13. Limitations

825 Sieve inherits the reach of lexical statistics, and the theory section states
826 the condition rather than hiding it: where evidence shares no vocabulary
827 with the question, 1 fails and the relevance term goes blind. Paraphrased
828 evidence scores poorly; multilingual inputs need language-matched stopword
829 lists and segmentation; numbers, citations, and code identifiers, which domi-
830 nate scientific text, are weighted only crudely by IDF. The margin analysis of
831 section 12 makes the dependence measurable, not mysterious, and the BM25
832 rows show the same dependence from the outside.

833 The evaluation has five boundaries. First, the end-task protocol covers
834 one reader (GLM-4-Plus) on one benchmark (QASPER) at one operating
835 point ($4\times$) with $n=25$: its intervals are wide, its claims are scoped to match-
836 ing the strongest cheap baseline, and reader-dependence is untested. An
837 expanded end-task protocol ($n = 100$, seven selectors including BM25, same
838 reader and operating point) is implemented and released with the repository,
839 and its execution was blocked during this study by sustained rate limits on
840 the reader’s public API rather than by design; the released script resumes
841 from checkpoints, and the end-task numbers will be updated in the artifact
842 without touching any selector-level claim. Second, the neural comparison
843 instantiates the perplexity-gate mechanism with GPT-2 small at sentence
844 granularity; it is the family’s cost floor and a faithful mechanism replica, but
845 it is not LLMLingua’s full coarse-to-fine machinery and not LLMLingua-2’s
846 distilled classifier, so the quality row of Table 4 is a lower bound on what the
847 neural corner can reach, and the cost row a lower bound as well. Third, the
848 reader-side positional profile that justifies the U-shaped prior is taken from

849 prior work [5] rather than re-measured under our reader; the selection-level
850 placement curve is measured, the reader-side one is borrowed. Fourth, the
851 recall–accuracy correlation of section 11 is estimated at $n = 25$ per method,
852 so its magnitude carries wide intervals even though its sign is stable across
853 methods. Fifth, sentence-level granularity bounds the useful compression
854 range: beyond $r \approx 16$ the output degenerates toward a few isolated sen-
855 tences, and token-level compressors retain an advantage at extreme ratios.

856 Two accounting caveats remain from the conference version. The reader
857 experiments measure accuracy under a fixed serving stack rather than wall-
858 clock serving of an open model, and the LLaMA-2 projection is analytical
859 with stated assumptions rather than a measured deployment. The gate’s
860 GPU floor is analytic and assumes a resident scorer at 40% utilization; a
861 hosted implementation with tokenization, budget iterations, and cold starts
862 costs more, and our CPU measurement of the gate is correspondingly the
863 like-for-like number for CPU-bound deployments.

864 14. Conclusion

865 We asked how much of long-context quality a compressor can keep with
866 no neural scorer at all, and answered it as a frontier study rather than a single
867 operating point. At the $4\times$ compression, Sieve keeps 57.9% of full-context
868 F1 end to end, statistically matched with the strongest cheap baseline, while
869 spending 53.7 ms of CPU time and zero model parameters. Under the reader-
870 free protocol, on 1600 evidence-annotated records across three benchmarks,
871 it retains more gold evidence than every question-agnostic selector with fully
872 separated intervals, and the margin over truncation grows with the compres-

873 sion ratio and with the strength of the question’s lexical signal, from 12.8
 874 points on abstract scientific questions to 44.8 on entity-named multi-hop
 875 ones. The added baselines price the remaining design space: BM25 shows
 876 that raw relevance ranking already recovers the most evidence at the least
 877 question-conditioned cost, which the selector’s own ablations explain: the full
 878 configuration knowingly trades that recall for positional reliability, and the
 879 measured GPT-2 perplexity-gate baseline, at its cost floor, buys no measur-
 880 able recall on these benchmarks, at three orders of magnitude more compute
 881 ($255\times$ at the 6,000-word operating point) and a hosted model. The formal
 882 analysis states when a stronger scorer should be expected to earn its infras-
 883 tructure: exactly when the question–evidence link stops being lexical. We
 884 see Sieve as the zero-cost first stage of a compression stack (composable with
 885 cache-level methods, auditable by construction, and tuned once and trans-
 886 ferred), and its measurements as a low-cost reference point against which
 887 more expensive context compressors can be evaluated.

888 **Declarations**

889 *Funding..* None.

890 *CRedit authorship contribution statement..* [**Author name withheld for**
 891 **double-anonymous review**]: Conceptualization, Methodology, Software,
 892 Investigation, Data curation, Writing – original draft, Writing – review &
 893 editing.

894 *Declaration of competing interest..* The authors declare that they have no
 895 known competing financial interests or personal relationships that could have
 896 appeared to influence the work reported in this paper.

897 *Data availability.* The benchmarks are public (QASPER, HotpotQA, 2Wiki-
898 MultihopQA). The released repository contains the pool-construction scripts
899 with fixed seeds, the sampled pools, every raw checkpoint, and one command
900 per table and figure, so that every number in this paper can be regenerated
901 from scratch.

902 References

- 903 [1] H. Jiang, Q. Wu, C.-Y. Lin, Y. Yang, L. Qiu, LlmLingua: Compress-
904 ing prompts for accelerated inference of large language models (2023)
905 1335813376doi:10.18653/v1/2023.emnlp-main.825.
906 URL <http://dx.doi.org/10.18653/v1/2023.emnlp-main.825>
- 907 [2] H. Jiang, Q. Wu, X. Luo, D. Li, C.-Y. Lin, Y. Yang, L. Qiu, Longllm-
908 lingua: Accelerating and enhancing llms in long context scenarios via
909 prompt compression (2024) 16581677doi:10.18653/v1/2024.acl-long
910 g.91.
911 URL <http://dx.doi.org/10.18653/v1/2024.acl-long.91>
- 912 [3] Z. Pan, Q. Wu, H. Jiang, M. Xia, X. Luo, J. Zhang, Q. Lin, V. Rühle,
913 Y. Yang, C.-Y. Lin, H. V. Zhao, L. Qiu, D. Zhang, LlmLingua-2: Data
914 distillation for efficient and faithful task-agnostic prompt compression
915 (2024) 963981doi:10.18653/v1/2024.findings-acl.57.
916 URL <http://dx.doi.org/10.18653/v1/2024.findings-acl.57>
- 917 [4] Y. Li, B. Dong, F. Guerin, C. Lin, Compressing context to enhance
918 inference efficiency of large language models (2023) 63426353doi:10.1

- 8653/v1/2023.emnlp-main.391.
 URL <http://dx.doi.org/10.18653/v1/2023.emnlp-main.391>
- [5] N. F. Liu, K. Lin, J. Hewitt, A. Paranjape, M. Bevilacqua, F. Petroni, P. Liang, Lost in the middle: How language models use long contexts (2023). doi:10.48550/ARXIV.2307.03172.
 URL <https://arxiv.org/abs/2307.03172>
- [6] J. Carbonell, J. Goldstein, The use of mmr, diversity-based reranking for reordering documents and producing summaries, in: Proceedings of the 21st annual international ACM SIGIR conference on Research and development in information retrieval, SIGIR98, ACM, 1998, p. 335336. doi:10.1145/290941.291025.
 URL <http://dx.doi.org/10.1145/290941.291025>
- [7] S. Robertson, H. Zaragoza, The probabilistic relevance framework: Bm25 and beyond, Foundations and Trends in Information Retrieval 4 (1-2) (2009) 1174. doi:10.1561/15000000019.
 URL <http://dx.doi.org/10.1561/15000000019>
- [8] P. Dasigi, K. Lo, I. Beltagy, A. Cohan, N. A. Smith, M. Gardner, A dataset of information-seeking questions and answers anchored in research papers, in: Proceedings of the 2021 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Association for Computational Linguistics, 2021, p. 45994610. doi:10.18653/v1/2021.naacl-main.365.
 URL <http://dx.doi.org/10.18653/v1/2021.naacl-main.365>

- 942 [9] Z. Yang, P. Qi, S. Zhang, Y. Bengio, W. Cohen, R. Salakhutdinov,
943 C. D. Manning, Hotpotqa: A dataset for diverse, explainable multi-hop
944 question answering, in: Proceedings of the 2018 Conference on Empiri-
945 cal Methods in Natural Language Processing, Association for Computa-
946 tional Linguistics, 2018, p. 23692380. doi:10.18653/v1/d18-1259.
947 URL <http://dx.doi.org/10.18653/v1/d18-1259>
- 948 [10] X. Ho, A.-K. D. Nguyen, S. Sugawara, A. Aizawa, Constructing a multi-
949 hop QA dataset for comprehensive evaluation of reasoning steps, in:
950 Proceedings of the 28th International Conference on Computational Lin-
951 guistics, 2020, pp. 6601–6607. doi:10.48550/arXiv.2011.01060.
- 952 [11] Z. Zhang, Y. Sheng, T. Zhou, T. Chen, L. Zheng, R. Cai, Z. Song,
953 Y. Tian, C. Ré, C. Barrett, Z. Wang, B. Chen, H₂O: Heavy-hitter oracle
954 for efficient generative inference of large language models (2023). doi:
955 10.48550/ARXIV.2306.14048.
956 URL <https://arxiv.org/abs/2306.14048>
- 957 [12] G. Xiao, Y. Tian, B. Chen, S. Han, M. Lewis, Efficient streaming lan-
958 guage models with attention sinks (2023). doi:10.48550/ARXIV.2309.
959 17453.
960 URL <https://arxiv.org/abs/2309.17453>
- 961 [13] T. Dao, D. Y. Fu, S. Ermon, A. Rudra, C. Ré, Flashattention: Fast
962 and memory-efficient exact attention with io-awareness (2022). doi:
963 10.48550/ARXIV.2205.14135.
964 URL <https://arxiv.org/abs/2205.14135>

- [14] W. Kwon, Z. Li, S. Zhuang, Y. Sheng, L. Zheng, C. H. Yu, J. E. Gonzalez, H. Zhang, I. Stoica, Efficient memory management for large language model serving with pagedattention (2023). doi:10.48550/ARXIV.2309.06180.
URL <https://arxiv.org/abs/2309.06180>
- [15] R. Mihalcea, P. Tarau, TextRank: Bringing order into text, in: D. Lin, D. Wu (Eds.), Proceedings of the 2004 Conference on Empirical Methods in Natural Language Processing, Association for Computational Linguistics, Barcelona, Spain, 2004, pp. 404–411.
URL <https://aclanthology.org/W04-3252/>
- [16] J. Kaplan, S. McCandlish, T. Henighan, T. B. Brown, B. Chess, R. Child, S. Gray, A. Radford, J. Wu, D. Amodei, Scaling laws for neural language models, arXiv preprint arXiv:2001.08361 (2020).
- [17] Y. Bai, X. Lv, J. Zhang, H. Lyu, J. Tang, Z. Huang, Z. Du, X. Liu, A. Zeng, L. Hou, Y. Dong, J. Tang, J. Li, LongBench: A bilingual, multitask benchmark for long context understanding, arXiv preprint arXiv:2308.14508 (2023).
- [18] H. Trivedi, N. Balasubramanian, T. Khot, A. Sabharwal, MuSiQue: Multihop questions via single-hop question composition, in: Transactions of the Association for Computational Linguistics, Vol. 10, 2022, pp. 539–554.
- [19] T. GLM, :, A. Zeng, B. Xu, B. Wang, C. Zhang, D. G. Yin, D. Zhang, D. Rojas, G. Feng, H. Zhao, H. Lai, H. Yu, H. Wang, J. Sun, J. Zhang,

988 J. Cheng, J. Gui, J. Tang, J. Zhang, J. Sun, J. Li, L. Zhao, L. Wu,
989 L. Zhong, M. Liu, M. Huang, P. Zhang, Q. Zheng, R. Lu, S. Duan,
990 S. Zhang, S. Cao, S. Yang, W. L. Tam, W. Zhao, X. Liu, X. Xiao,
991 X. Zhang, X. Gu, L. Xin, X. Liu, X. Liu, X. Yang, X. Song, X. Zhang,
992 Y. An, Y. Xu, Y. Niu, Y. Yang, Y. Li, Y. Bai, Y. Dong, Z. Qi,
993 Z. Wang, Z. Yang, Z. Du, Z. Hou, Z. Wang, Chatglm: A family of
994 large language models from glm-130b to glm-4 all tools, arXiv preprint
995 arXiv:2406.12793 (2024).

996 [20] A. Radford, J. Wu, R. Child, D. Luan, D. Amodei, I. Sutskever, Lan-
997 guage models are unsupervised multitask learnersOpenAI technical re-
998 port. [https://cdn.openai.com/better-language-models/language](https://cdn.openai.com/better-language-models/language_models_are_unsupervised_multitask_learners.pdf)
999 [_models_are_unsupervised_multitask_learners.pdf](https://cdn.openai.com/better-language-models/language_models_are_unsupervised_multitask_learners.pdf) (2019).

1000 [21] H. Touvron, L. Martin, K. Stone, P. Albert, A. Almahairi, Y. Babaei,
1001 N. Bashlykov, S. Batra, P. Bhargava, S. Bhosale, D. Bikel, L. Blecher,
1002 C. C. Ferrer, M. Chen, G. Cucurull, D. Esiobu, J. Fernandes, J. Fu,
1003 W. Fu, B. Fuller, C. Gao, V. Goswami, N. Goyal, A. Hartshorn, S. Hos-
1004 seini, R. Hou, H. Inan, M. Kardas, V. Kerkez, M. Khabsa, I. Kloumann,
1005 A. Korenev, P. S. Koura, M.-A. Lachaux, T. Lavril, J. Lee, D. Liskovich,
1006 Y. Lu, Y. Mao, X. Martinet, T. Mihaylov, P. Mishra, I. Molybog,
1007 Y. Nie, A. Poulton, J. Reizenstein, R. Rungta, K. Saladi, A. Schel-
1008 ten, R. Silva, E. M. Smith, R. Subramanian, X. E. Tan, B. Tang,
1009 R. Taylor, A. Williams, J. X. Kuan, P. Xu, Z. Yan, I. Zarov, Y. Zhang,
1010 A. Fan, M. Kambadur, S. Narang, A. Rodriguez, R. Stojnic, S. Edunov,
1011 T. Scialom, Llama 2: Open foundation and fine-tuned chat models

1012 (2023). doi:10.48550/ARXIV.2307.09288.

1013 URL <https://arxiv.org/abs/2307.09288>

1014 **Appendix A. Implementation details**

1015 Sentence segmentation splits on punctuation boundaries with guards for
1016 common abbreviations (*et al.*, *e.g.*, *Fig.*) and merges fragments that end
1017 in a guarded token; the segmenter is deterministic. Tokenization lowercases
1018 and keeps alphanumeric and hyphenated terms; stopwords are a fixed 150-
1019 word English list included with the code. The conference-version experiments
1020 ran on one core of a container with 8 vCPUs and 32 GB of memory; the
1021 journal-version experiments, including the GPT-2 gate (int8, two threads)
1022 and every timing number in section 9, ran in a container with 2 vCPUs
1023 and 4 GB of memory, so that all same-hardware comparisons are like-for-
1024 like. Timing figures are medians of five runs after one warm-up (three for
1025 the gate). The full pipeline (segmentation, scoring, selection) is a single
1026 Python module with NumPy as its only dependency beyond the standard
1027 library; the gate baseline additionally uses Transformers and PyTorch. The
1028 repository reproduces every table and figure from the raw checkpoints with
1029 one command per artifact, and the macro file that carries every number in
1030 this paper into its LaTeX source is itself generated from those checkpoints,
1031 so no number in the text was typed by hand.

1032 **Appendix B. Proofs and formal statements**

1033 *Proof of 1.* Each pipeline stage touches each token a constant number of
1034 times: segmentation and tokenization are single passes over the W words; the

1035 IDF document-frequency table is one accumulation pass over sentence term
1036 sets; relevance evaluates each sentence against the query term set in $O(|s_i| +$
1037 $|Q|)$; the prior is $O(n)$. The greedy loop performs at most k iterations, each
1038 scanning the $O(n)$ remaining candidates and evaluating the marginal penalty
1039 against the $O(k)$ selected sentences at $O(\bar{s})$ per pair on sparse counters, giving
1040 $O(k n \bar{s})$ worst case; peak memory stays at the token counters because the
1041 released implementation recomputes similarities rather than memoizing them.

1042

□

1043 *Proof of 2.* A forward pass of a decoder-only transformer with N parameters
1044 over L tokens costs approximately $2NL$ FLOPs for the matrix multiplications
1045 that dominate the arithmetic [16]; any scorer-based compressor computes at
1046 least one such pass over the full context before it can rank its units, and
1047 coarse-to-fine budget searches compute one per iteration. The scorer must
1048 be resident, so $c_{\text{host}} \neq 0$.

□

1049 *Proof of 3.* Greedy selection under $\alpha = 1$, flat prior, and $\beta = 0$ orders sen-
1050 tences by rel alone. Under 1, for any $e \in E$ and $s \in C \setminus E$, $\text{rel}(e) \geq \text{rel}(s)$,
1051 so every evidence sentence is admitted before every dominated non-evidence
1052 sentence; ties among evidence sentences do not affect the claim because the
1053 budget condition $B \geq \sum_{e \in E} |e|$ admits all of E whenever they are reached,
1054 and they are reached before the budget can be exhausted by non-evidence
1055 sentences that follow them in the order. Hence $\rho = 1$.

□

1056 *Proof of 4.* Truncation keeps the prefix of $\lceil n/r \rceil$ sentences. For an evidence
1057 sentence at a uniform position, the retention probability is $\lceil n/r \rceil / n \in [1/r, 1/r +$

1058 $1/n]$, and independence across evidence sentences plus linearity of expecta-
1059 tion gives $\mathbb{E}[\rho] = \min(1, 1/r) + O(1/n)$. $\square$

1060 **Appendix C. Additional results**

1061 *Compute accounting..* The reader evaluation consumed 256 completed API
1062 calls across the development grid and the main comparison; the journal ver-
1063 sion added no reader calls. The journal-version selector-protocol runs covered
1064 1600 records across three pools with roughly forty selector invocations per
1065 record (methods, sweeps, ablations, sensitivity); the GPT-2 gate ran 100
1066 QASPER records at 7.5s median scoring each. Compression-side compute
1067 for all journal experiments was under 6 vCPU-hours including the gate; no
1068 GPUs were used for any measurement reported in this paper.

1069 *Development grid..* The development grid over $\alpha \in \{0.3, 0.6, 0.9\}$ at $r = 4$
1070 gave dev F1 of 52.6, 47.1, and 50.4 respectively; the differences are within
1071 the noise of a 20-example development split, and we selected $\alpha = 0.3$ for the
1072 best dev F1. The dev and test numbers are not directly comparable: the
1073 development split mixes ten QASPER records with ten HotpotQA distrac-
1074 tor records, and the multi-paragraph HotpotQA items are far easier for the
1075 reader, which inflates the dev mean relative to the QASPER-only test split.
1076 The α choice is a point estimate from a noisy grid, and the paper’s claims
1077 do not rest on it being optimal.

1078 *Conference-version compatibility..* The conference version’s 191-record QASPER
1079 pool, its timing host, and its numbers remain reproducible from the original
1080 artifacts; the journal pools are supersets drawn under the same filters, and

1081 every qualitative conference claim (ordering of methods, growth of the mar-
1082 gin in r , component attribution) reproduces at the larger scale with tighter
1083 intervals. Three conference-version statements are superseded: the QASPER
1084 recall intervals, which touched at their edges at $n = 191$, are separated at
1085 $n=1000$; the neural-baseline comparison, which was analytic in the confer-
1086 ence version, is measured here; and the baseline set now includes BM25,
1087 which the conference version lacked entirely.